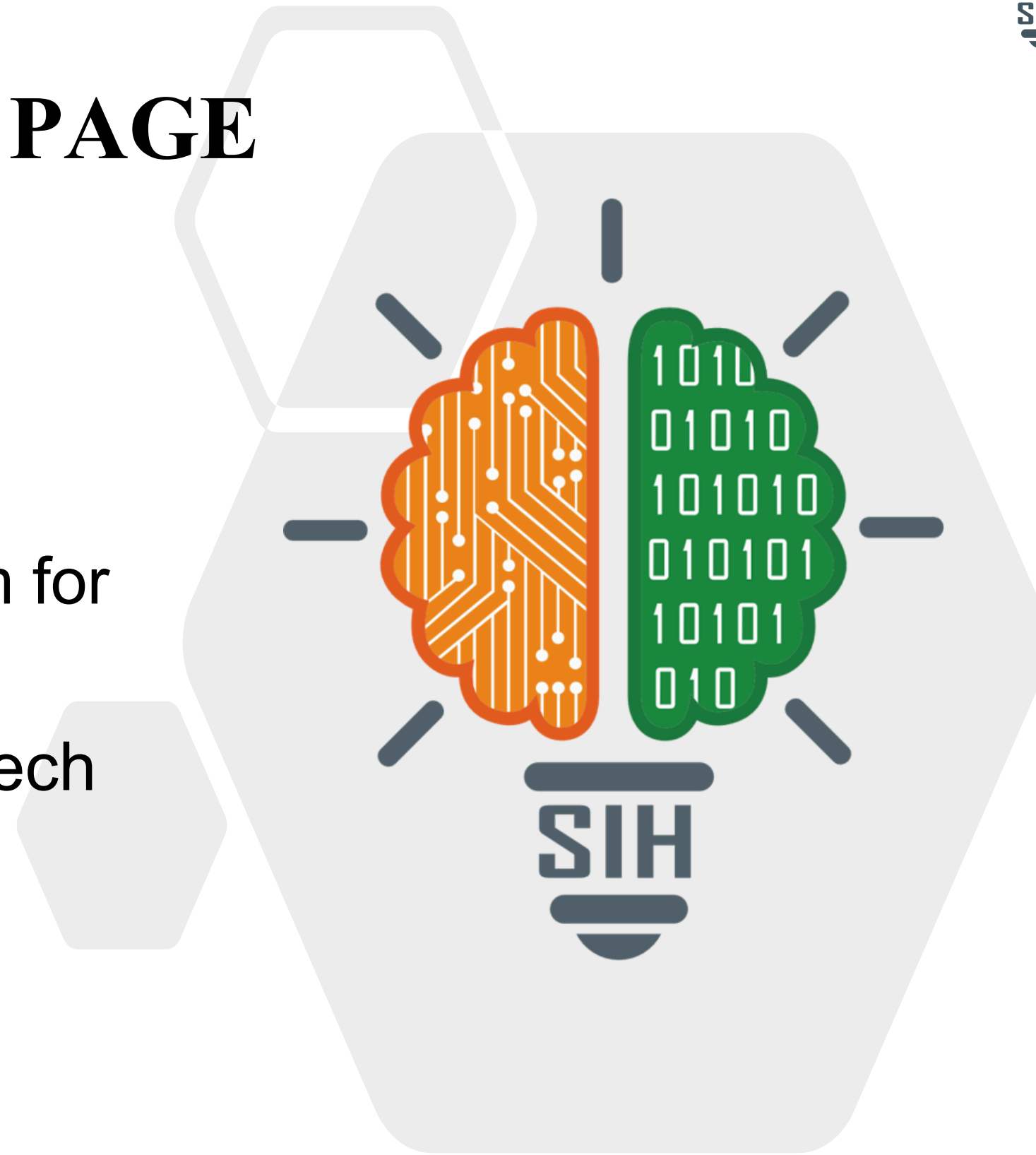
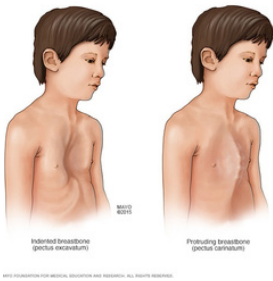


TITLE PAGE

- **Problem Statement ID** –26139
- **Problem Statement Title**-Hybrid Quantum Machine Learning Platform for Early Disease Detection
- **Theme**-MedTech/BioTech/HealthTech
- **PS Category**- Software
- **Team ID**-Tanvi8810506680
- **Team Name**:Kasukabe.exe



Hybrid quantum-classical diagnosis of rare disease like alike



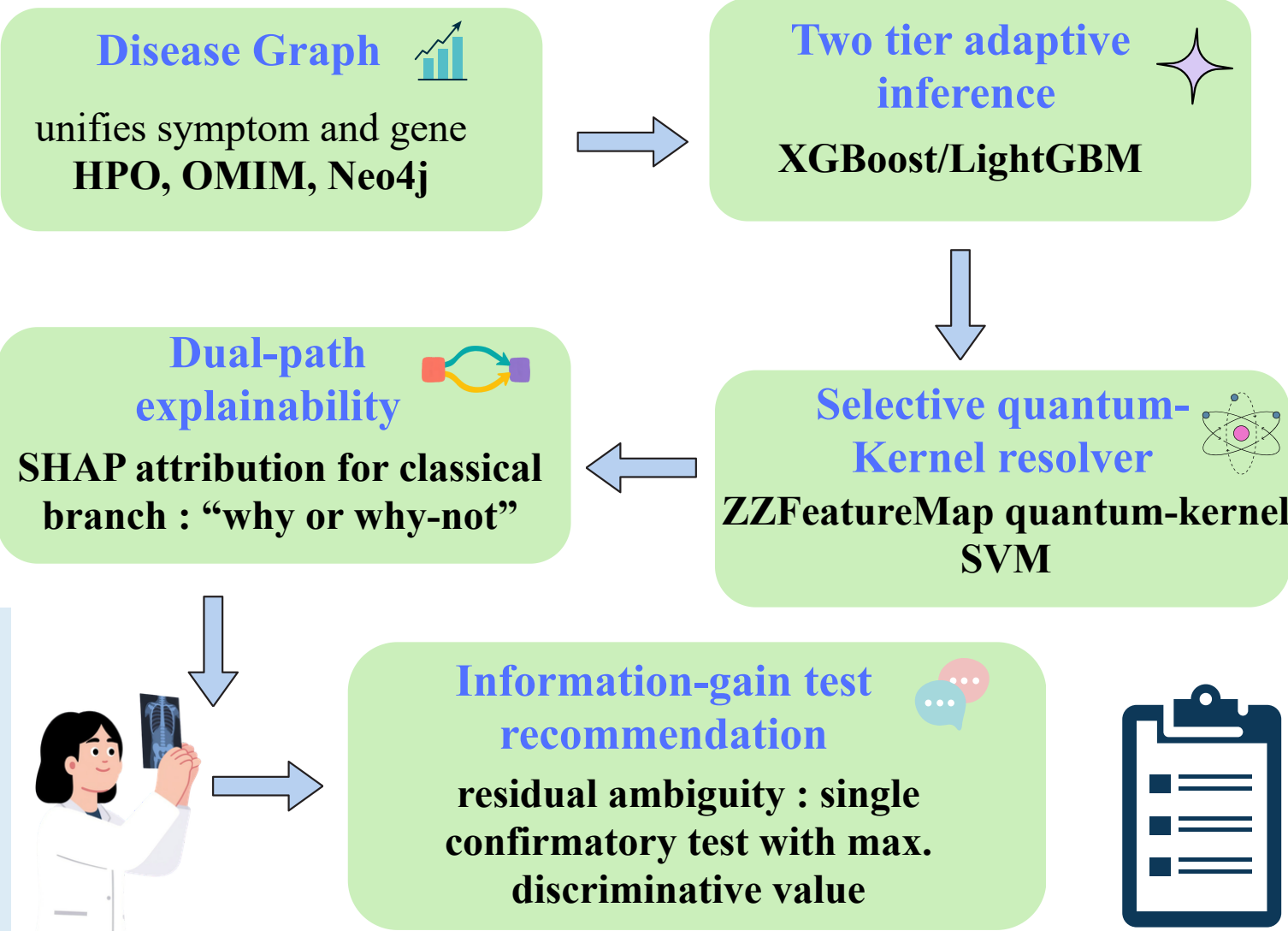
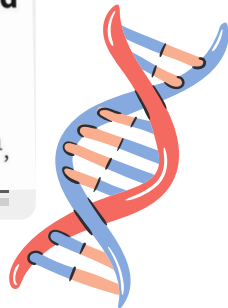
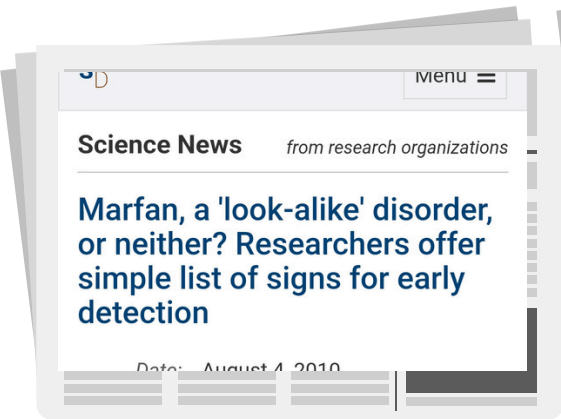
Marfan syndrome is an inherited disorder that damages the body's connective tissue and affects multiple organ systems.

- **Gene : FBN1**

Condition	Gene(s)	Shared with Marfan
Loeys-Dietz syndrome	TGFBR1, TGFBR2, SMAD3, TGFB2	Aortic disease, long fingers, sunken chest, spine curve
Beals syndrome	FBN2	Marfan-like body shape, long fingers
Shprintzen-Goldberg syndrome	SKI	Marfan-like skeleton, spine curve

- Drawbacks**
- **Phenotypic overlap** across genotypically distinct disorders
 - Shared **pathway etiology** masking genetic divergence
 - **Label-disjoint** model architecture
 - Absence of calibrated **ambiguity** quantification
 - Lack of **counterfactual/contrastive** explainability
 - No **decision-theoretic** next-step recommendation

QResolve is a hybrid quantum-classical diagnostic assistant that reframes rare-disease diagnosis from single-label classification to **confusion resolution** - identifying not just the most probable disease, but the evidence that rules out its closest phenotypic rival.



U S P

Compute-Proportional Quantum Use

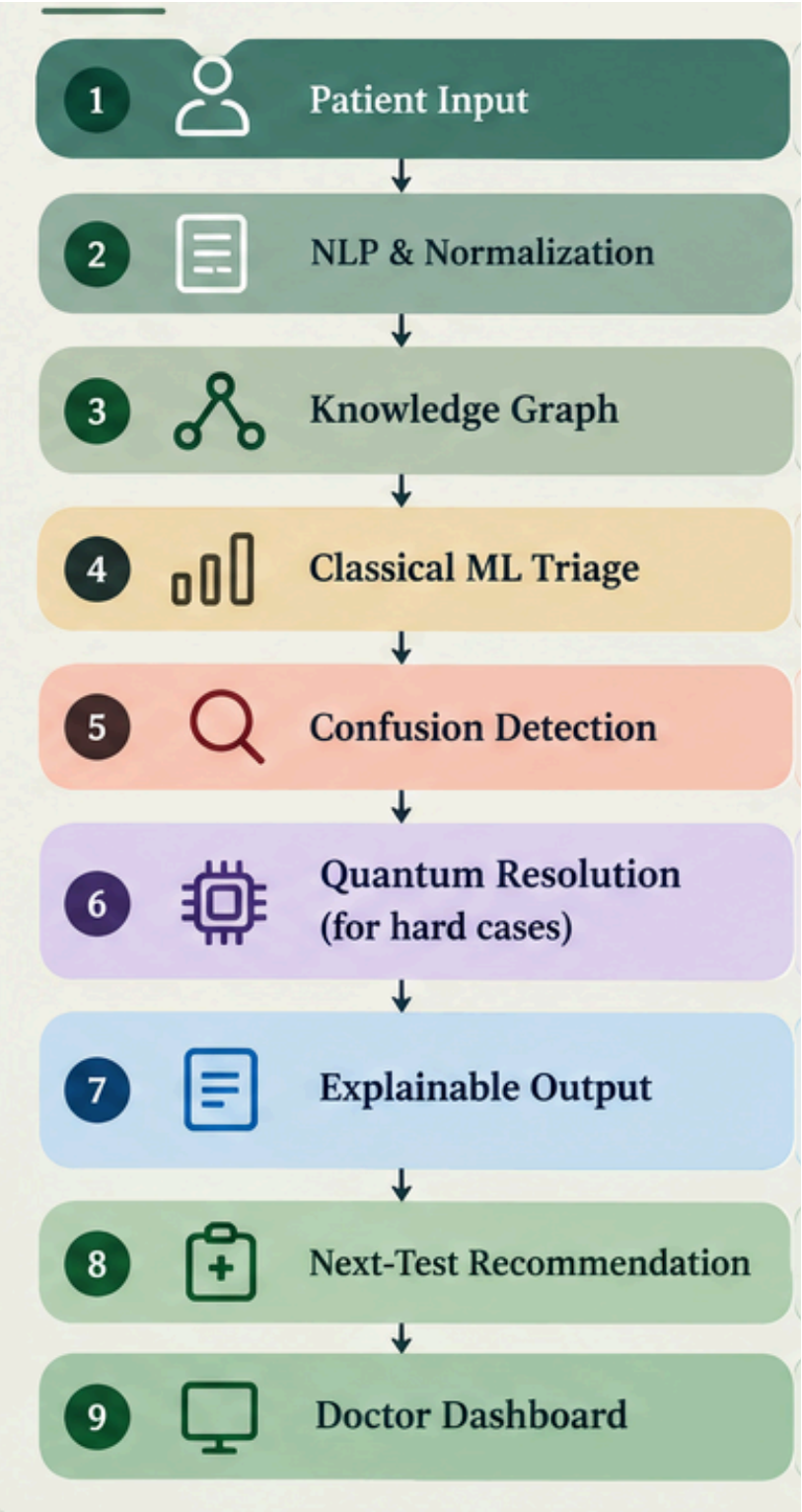
Decision-Theoretic Closure

Contrastive Explainability

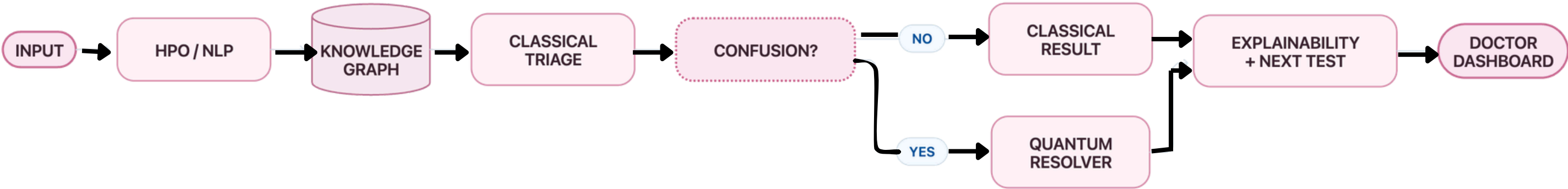
Falsifiable Quantum-Advantage Claim

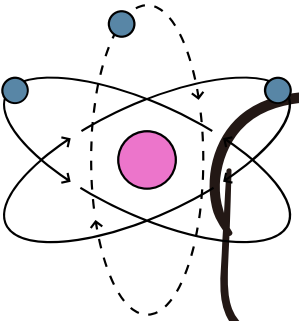
Confusion-Resolution Paradigm

Methodology



System Architecture





Project Feasibility

Data Feasibility

Required disease, symptom and genetic information can be sourced from established biomedical databases.

Model Feasibility

The ZZFeatureMap quantum kernel SVM avoids iterative training, making it suitable for a small-scale 8–16 qubit benchmark.

Simulator-Based Implementation

Qiskit Aer enables quantum-model development and testing without physical quantum hardware.

Potential Challenges And Risk

Rare-Disease Data Scarcity

Limited labeled hard-case data makes training the confusion resolver challenging.

Unproven Quantum Advantage

The quantum kernel may not outperform the classical model on hard cases.

NISQ Hardware Noise & Simulator Gap

Real quantum hardware is noisy, and simulator results may differ from actual devices.

Risk Mitigation Approach

Focused Dataset

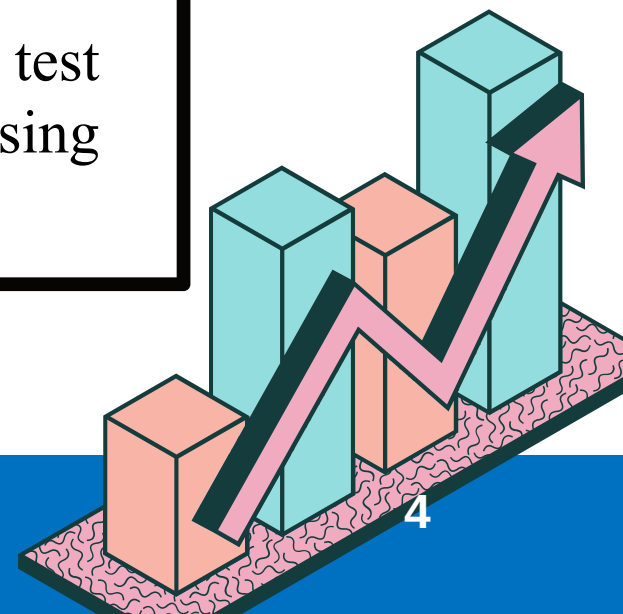
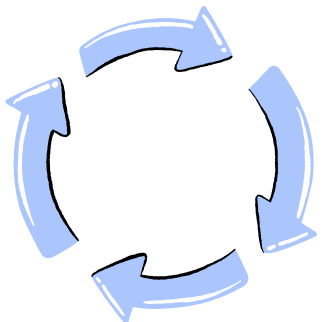
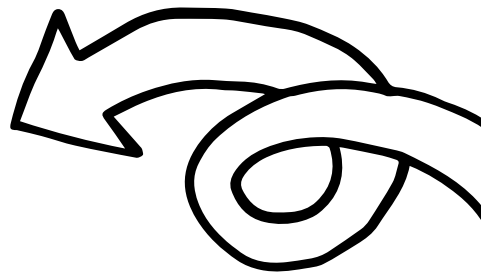
Start with a focused Marfan-like disease cluster and use curated HPO, Orphanet, OMIM and ClinVar data.

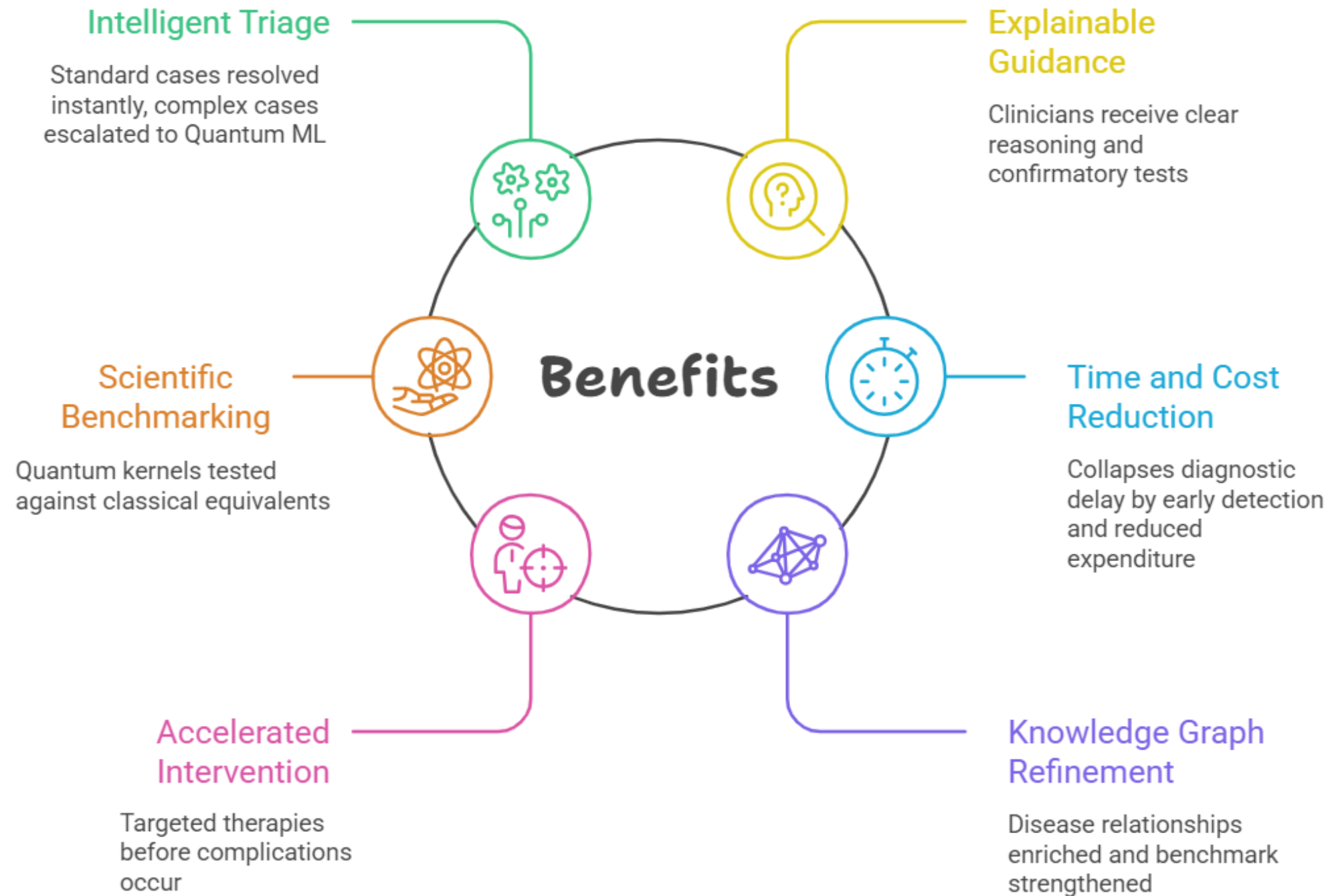
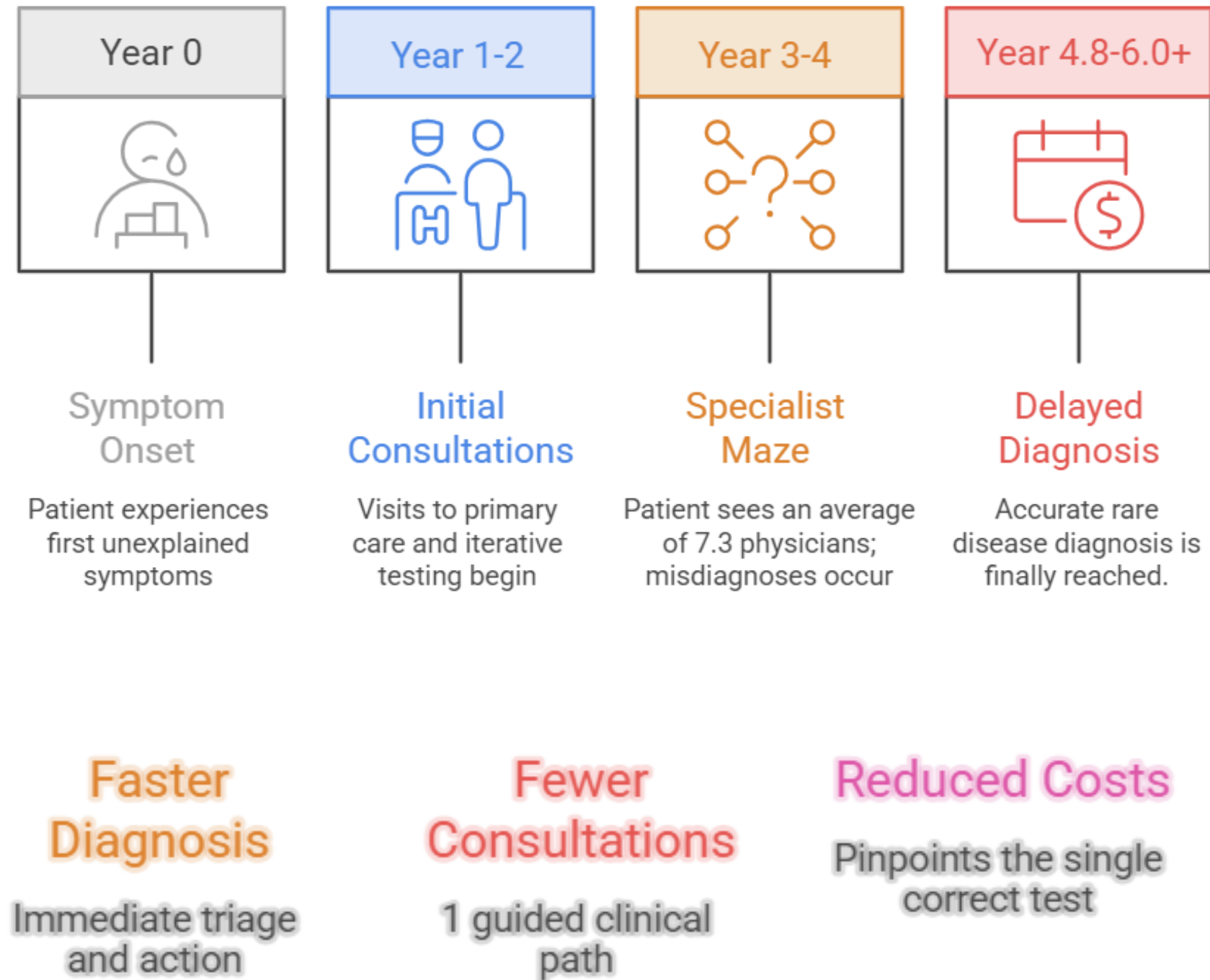
Fair Quantum Benchmarking

Compare Quantum Kernel SVM with classical RBF-SVM on the same hard cases.

Simulation-First Approach

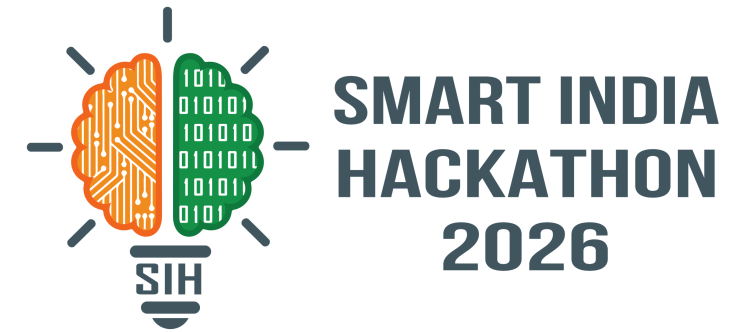
Use Qiskit Aer to develop and test the quantum model before using real quantum hardware.







RESEARCH AND REFERENCES



- <https://pubmed.ncbi.nlm.nih.gov/39015450/>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC6826414/>
- The Journal of Rare Disorders via Global Genes; The R69 Initiative on Diagnostic Uncertainty.
- <https://www.ncbi.nlm.nih.gov/clinvar/search/>
- Global Genes survey of patients and healthcare professionals.